

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2003

Spearfish Hydro 100 kV
March 04, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2003. The Interconnection Customer proposes a 130.3 MW solar plus storage facility interconnecting at the Spearfish Hydro 100 kV substation in the Illinois area. The request was received on February 08, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 453 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,588,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SPEARFISH HYDRO to ARCH SMITH 100 kV Circuit 1
Contingency: ARCH SMITH to SPEARFISH HYDRO 100 kV Ckt 2
Loading: 121.5% (Rating: 218 MVA)
DFAX: 30.8%
Pre-project loading: 103.1%
2. Facility: SPEARFISH HYDRO to ARCH SMITH 100 kV Circuit 2
Contingency: ARCH SMITH to SPEARFISH HYDRO 100 kV
Loading: 121.5% (Rating: 218 MVA)
DFAX: 30.8%
Pre-project loading: 103.1%
3. Facility: SWATT to LAWRENCE HILL 100 kV
Contingency: S BOSTON to BLYTHE 230 kV line
Loading: 106.4% (Rating: 392 MVA)
DFAX: 29.6%
Pre-project loading: 96.6%
4. Facility: BLYTHE 230/100kV Transformer Bank Bank 1
Contingency: N BONNEVILLE to LAKE CHARLES PLANT 100 kV
Loading: 104.8% (Rating: 242 MVA)
DFAX: 14.7%
Pre-project loading: 96.9%
5. Facility: BLYTHE 230-100 kV Auto Bank 2
Contingency: LAKE CHARLES PLANT to NORTH BONNEVILLE 100 kV line
Loading: 104.8% (Rating: 242 MVA)
DFAX: 14.7%
Pre-project loading: 96.9%

6. Facility: BLYTHE 230 kV/100 kV Transformer
 Contingency: NORTH BONNEVILLE to LAKE CHARLES PLANT 100 kV
 Loading: 104.8% (Rating: 242 MVA)
 DFAX: 14.7%
 Pre-project loading: 96.9%
7. Facility: TINLEY PARK to SWEATT 100 kV
 Contingency: SWEATT to LAWRENCE HILL 100 kV
 Loading: 104.6% (Rating: 273 MVA)
 DFAX: 25.6%
 Pre-project loading: 92.4%
8. Facility: TERRYVILLE to LAWRENCE HILL 100 kV
 Contingency: BLYTHE to S BOSTON 230 kV Line
 Loading: 102.1% (Rating: 133 MVA)
 DFAX: 10.2%
 Pre-project loading: 92.1%
9. Facility: BLYTHE to SOUTH BOSTON 230 kV line
 Contingency: N BONEVILLE SUBSTATION to LAKE CHARLES PLANT 100 kV
 Loading: 101.0% (Rating: 849 MVA)
 DFAX: 15.4%
 Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SPEARFISH HYDRO to ARCH SMITH 100 kV	Rebuild line with higher-capacity conductor	\$2,493,000	28% (\$693,000)
ARCH SMITH to SPEARFISH HYDRO 100 kV Line 2	Rebuild line with higher-capacity conductor	\$3,718,000	28% (\$1,034,000)
SWATT to LAWRENCE HILL 100 kV	Reconductor line	\$892,000	16% (\$138,000)
BLYTHE 230/100kV Transformer Bank 1	Replace transformer with higher-rated unit	\$11,931,000	6% (\$656,000)
BLYTHE 230-100 kV Auto Bank 2	Replace transformer with higher-rated unit	\$7,136,000	6% (\$393,000)
BLYTHE SUBSTATION 230/100kV Transformer	Replace transformer with higher-rated unit	\$9,557,000	6% (\$526,000)
TINLEY PARK to SWEATT 100 kV	Reconductor line	\$2,179,000	17% (\$377,000)
TERRYVILLE to LAWRENCE HILL 100 kV	Reconductor line	\$600,000	60% (\$358,000)
BLYTHE to SOUTH BOSTON 230 kV line	Reconductor line	\$10,420,000	4% (\$413,000)
	Total allocated to project		\$4,588,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.